

# Notes: Caballero, Farhi, and Gourinchas (AER, 2008)

## An Equilibrium Model of “Global Imbalances” and Low Interest Rates

### Contents

|                                                                              |          |
|------------------------------------------------------------------------------|----------|
| <b>Notes: Caballero, Farhi, and Gourinchas (AER, 2008)</b>                   | <b>1</b> |
| An Equilibrium Model of “Global Imbalances” and Low Interest Rates . . . . . | 1        |
| <b>1 One-sentence summary</b>                                                | <b>2</b> |
| <b>2 Big picture</b>                                                         | <b>2</b> |
| 2.1 The three facts to be explained . . . . .                                | 2        |
| 2.2 Asset shortage rather than only saving glut . . . . .                    | 2        |
| 2.3 The role of the United States . . . . .                                  | 2        |
| 2.4 What the model jointly explains . . . . .                                | 2        |
| 2.5 Why the model matters . . . . .                                          | 2        |
| <b>3 Incremental contribution of the paper</b>                               | <b>2</b> |
| 3.1 A unified equilibrium explanation of three anomalies . . . . .           | 2        |
| 3.2 Asset-supply heterogeneity as the primitive . . . . .                    | 2        |
| 3.3 A formal theory of global asset scarcity . . . . .                       | 3        |
| 3.4 A framework for Asian crisis and emerging-market integration . . . . .   | 3        |
| 3.5 Extensions to FDI and exchange rates . . . . .                           | 3        |
| 3.6 Foundation for later safe-asset literature . . . . .                     | 3        |
| <b>4 Conceptual framework</b>                                                | <b>3</b> |
| 4.1 Regions and asset-producing capacity . . . . .                           | 3        |
| 4.2 Asset supply . . . . .                                                   | 3        |
| 4.3 Asset demand . . . . .                                                   | 3        |
| 4.4 Autarky interest rate . . . . .                                          | 4        |
| 4.5 Global equilibrium . . . . .                                             | 4        |
| <b>5 Model setup and key mechanisms</b>                                      | <b>4</b> |
| 5.1 Small open economy logic . . . . .                                       | 4        |
| 5.2 Two-region world . . . . .                                               | 4        |
| 5.3 Permanent decline in asset supply . . . . .                              | 4        |
| 5.4 Emerging-market growth . . . . .                                         | 4        |
| 5.5 FDI extension . . . . .                                                  | 4        |
| 5.6 Two-good exchange-rate extension . . . . .                               | 5        |
| <b>6 Calibration and quantitative exercises</b>                              | <b>5</b> |
| 6.1 Main calibration . . . . .                                               | 5        |

|           |                                                                                                                            |           |
|-----------|----------------------------------------------------------------------------------------------------------------------------|-----------|
| 6.2       | Asian crisis exercise . . . . .                                                                                            | 5         |
| 6.3       | Emerging trends exercise . . . . .                                                                                         | 5         |
| 6.4       | Interpretation of quantitative fit . . . . .                                                                               | 5         |
| <b>7</b>  | <b>Identification and interpretation</b>                                                                                   | <b>5</b>  |
| 7.1       | What the paper identifies theoretically . . . . .                                                                          | 5         |
| 7.2       | How stylized facts discipline the model . . . . .                                                                          | 5         |
| 7.3       | Why the mechanism is distinct from standard views . . . . .                                                                | 5         |
| 7.4       | Limitations . . . . .                                                                                                      | 6         |
| 7.5       | Observable implications . . . . .                                                                                          | 6         |
| <b>8</b>  | <b>Tables and figures</b>                                                                                                  | <b>6</b>  |
| 8.1       | Figure 1: Three Stylized Facts . . . . .                                                                                   | 6         |
| 8.2       | Figure 2: The Metzler Diagram and Figure 3: The Metzler Diagram for a Permanent Drop in $\delta^R$ . . . . .               | 7         |
| 8.3       | Figure 4: Permanent Collapse in $\delta^R$ and Table 1: Main Parameters . . . . .                                          | 8         |
| 8.4       | Table 2: Calibrated Exercise 1: Asian Shocks and Table 3: Calibrated Exercise 2: Emerging Trends . . . . .                 | 9         |
| 8.5       | Figure 5: Collapse in $\delta^R$ with and without FDI and Figure 6: Collapse in $\delta^R$ in the Two-Good Model . . . . . | 9         |
| <b>9</b>  | <b>Short summary</b>                                                                                                       | <b>10</b> |
| <b>10</b> | <b>Conclusion</b>                                                                                                          | <b>11</b> |
| 10.1      | Core conclusion . . . . .                                                                                                  | 11        |
| 10.2      | Economic intuition . . . . .                                                                                               | 11        |
| 10.3      | Incremental contribution revisited . . . . .                                                                               | 11        |
| 10.4      | Implications . . . . .                                                                                                     | 11        |
| 10.5      | Limitations . . . . .                                                                                                      | 11        |
| 10.6      | Takeaway . . . . .                                                                                                         | 11        |

# 1 One-sentence summary

Caballero, Farhi, and Gourinchas build a global-equilibrium asset-scarcity model in which countries differ in their ability to generate financial assets, explaining persistent U.S. current account deficits, falling world real interest rates, and the rising global portfolio share of U.S. assets as joint equilibrium outcomes.

## 2 Big picture

### 2.1 The three facts to be explained

The paper begins from three empirical facts: the sustained U.S.-centered current account deficit, the decline in long-run real interest rates, and the growing share of U.S. assets in the portfolios of the rest of the world. The key idea is that these facts should be studied together rather than as separate anomalies.

### 2.2 Asset shortage rather than only saving glut

The mechanism is not simply that some countries save a lot. The deeper friction is that some regions cannot transform domestic income into reliable financial assets. When a region has high saving needs but low asset-creation capacity, its residents demand foreign assets.

### 2.3 The role of the United States

The United States, together with similar economies such as the United Kingdom and Australia, is modeled as an economy with a relatively high ability to produce pledgeable, store-of-value assets. This makes U.S. assets a global savings vehicle.

### 2.4 What the model jointly explains

The same asset-demand pressure can produce three outcomes simultaneously: capital flows into U.S. assets, current account deficits in the asset-producing region, and low world interest rates. The model therefore links global imbalances and low rates in one equilibrium system.

### 2.5 Why the model matters

The paper provides a general-equilibrium framework for thinking about global imbalances, financial development, safe-asset scarcity, foreign portfolio demand, and the limits of exchange-rate adjustment.

## 3 Incremental contribution of the paper

### 3.1 A unified equilibrium explanation of three anomalies

The paper's first contribution is to explain current account deficits, low interest rates, and rising foreign demand for U.S. assets in one model. Earlier discussions often treated these as separate puzzles.

### 3.2 Asset-supply heterogeneity as the primitive

The key primitive is countries' heterogeneous capacity to produce financial assets. This shifts the focus from preferences alone to the supply side of global store-of-value assets.

### 3.3 A formal theory of global asset scarcity

The paper formalizes the idea that emerging markets may grow quickly and save heavily while still lacking reliable domestic assets. This creates global demand for assets produced elsewhere.

### 3.4 A framework for Asian crisis and emerging-market integration

The quantitative exercises show that shocks to asset supply in Japan and emerging Asia, together with the integration of fast-growing emerging markets, can reproduce key features of the data.

### 3.5 Extensions to FDI and exchange rates

The paper extends the baseline asset-scarcity mechanism to foreign direct investment and two-good exchange-rate settings, showing which channels change the results and which leave the main mechanism intact.

### 3.6 Foundation for later safe-asset literature

The paper is an important predecessor of later work on safe-asset shortages, global imbalances, and the macro-financial role of the U.S. in supplying collateral-like assets to the world.

## 4 Conceptual framework

### 4.1 Regions and asset-producing capacity

The model divides the world into regions. A region's output grows, but only a fraction of its future output can be capitalized into financial assets. Let  $\delta^i$  denote the share of future output in region  $i$  that can be pledged and transformed into financial wealth.

### 4.2 Asset supply

In a small open economy with output  $X_t$  growing at rate  $g$ , the asymptotic value of capitalizable output is

$$\frac{V_t}{X_t} \rightarrow \frac{\delta}{r - g}.$$

Asset supply is decreasing in the interest rate because a higher discount rate lowers the value of future pledgeable output.

### 4.3 Asset demand

The demand for assets, normalized by output, is

$$\frac{W_t}{X_t} \rightarrow \frac{1 - \delta}{g + \theta - r}.$$

Here  $\theta$  governs the propensity to consume or the turnover rate of wealth. Asset demand is increasing in the interest rate because higher returns make saving more attractive.

## 4.4 Autarky interest rate

Equating asset supply and demand gives the autarky interest rate:

$$r_{aut} = g + \delta\theta.$$

A country with low  $\delta$  has a low autarky interest rate because it cannot generate enough domestic assets relative to domestic savings demand.

## 4.5 Global equilibrium

When regions integrate financially, a common world interest rate clears global asset markets. Regions with low autarky rates demand assets from regions with high autarky rates. This creates capital flows and persistent current account imbalances.

# 5 Model setup and key mechanisms

## 5.1 Small open economy logic

If the world interest rate  $r$  is below a region's autarky rate  $r_{aut}$ , domestic asset supply exceeds demand and the region runs a current account deficit. If  $r$  is above  $r_{aut}$ , the region runs a current account surplus.

$$\frac{CA_t}{X_t} \rightarrow -g \frac{r_{aut} - r}{(g + \theta - r)(r - g)}.$$

**Interpretation:** The sign of the current account is driven by the gap between the world interest rate and the domestic autarky rate. Asset-rich countries export assets and import goods, producing current account deficits.

## 5.2 Two-region world

The baseline two-region model compares  $U$ , the asset-producing region, with  $R$ , a region that has lower capacity to produce financial assets. A collapse in  $\delta^R$  reduces global asset supply and shifts world savings toward  $U$  assets.

## 5.3 Permanent decline in asset supply

A permanent fall in  $\delta^R$  lowers the world interest rate. It also increases the value of  $U$  assets, raises the share of  $U$  assets in global portfolios, and creates a persistent current account deficit in  $U$ .

## 5.4 Emerging-market growth

When the low-asset-supply region grows quickly, its demand for assets rises over time. If its asset-production capacity does not rise proportionally, its demand for foreign assets increases, deepening global imbalances.

## 5.5 FDI extension

FDI allows the asset-producing region to transform foreign projects into assets with higher pledgeability. This can reduce the long-run imbalance by improving global asset supply, but the baseline current-account mechanism remains important.

## 5.6 Two-good exchange-rate extension

The two-good model adds real exchange-rate adjustment. Exchange-rate movements help rebalance goods demand, but they do not eliminate the core asset-market mechanism driving low interest rates and global asset demand.

# 6 Calibration and quantitative exercises

## 6.1 Main calibration

The model is calibrated with a small set of parameters, including  $\theta$ ,  $g$ ,  $\delta$ , initial region output shares, and initial portfolio holdings. The aim is not high-frequency forecasting but order-of-magnitude discipline.

## 6.2 Asian crisis exercise

The first exercise treats the late-1990s Asian crisis as a negative shock to asset supply in the emerging region. The model produces a U.S. current account deficit, a decline in the world real interest rate, and a rise in the global share of U.S. assets.

## 6.3 Emerging trends exercise

The second exercise emphasizes the rise of fast-growing emerging markets with high savings needs and limited asset production capacity. Integration of these economies into global markets pushes down interest rates and raises demand for U.S.-type assets.

## 6.4 Interpretation of quantitative fit

The model is stylized and sometimes exaggerates movements because it abstracts from frictions such as liquidity, adjustment costs, and gradual market integration. The purpose is to show that the mechanism can generate movements of the right sign and approximate magnitude.

# 7 Identification and interpretation

## 7.1 What the paper identifies theoretically

The paper does not estimate a causal effect in the modern empirical sense. It identifies a mechanism: heterogeneity in asset-producing capacity can generate global capital flows, low interest rates, and portfolio shifts in equilibrium.

## 7.2 How stylized facts discipline the model

The model is disciplined by the joint behavior of current accounts, interest rates, and global portfolios. A successful model must explain all three together. Explaining only the U.S. current account without explaining low rates and portfolio shares is incomplete.

## 7.3 Why the mechanism is distinct from standard views

The standard view often explains deficits using U.S. consumption or fiscal behavior. The Caballero-Farhi-Gourinchas mechanism instead emphasizes foreign demand for U.S. assets and the scarcity of reliable financial

instruments abroad.

## 7.4 Limitations

The model abstracts from many features: nominal frictions, short-run fire sales, banking crises, heterogeneous households, institutional details of safe assets, and endogenous financial development. The calibrated simulations should be read as mechanism exercises, not precise forecasting tools.

## 7.5 Observable implications

The model predicts that global imbalances should be larger when fast-growing saving regions have limited asset creation, when asset-market collapses reduce pledgeable wealth abroad, and when U.S.-type assets retain high global collateral value.

# 8 Tables and figures

## 8.1 Figure 1: Three Stylized Facts

**Read:** The figure documents the three empirical facts that motivate the paper: persistent U.S.-centered current account deficits, falling world and U.S. real interest rates, and the rising share of U.S. assets in the rest of the world’s financial wealth and output.

**Economic message:** The paper asks for a model that explains all three patterns at once. The figure is therefore not merely descriptive; it is the target set of facts for the theory.

**Interpretation:** The important visual pattern is the co-movement. U.S. external deficits grow while real rates fall, which is hard to reconcile with a simple story of excessive U.S. borrowing demand. At the same time, the rest of the world increases its exposure to U.S. assets, consistent with a foreign demand for stores of value.

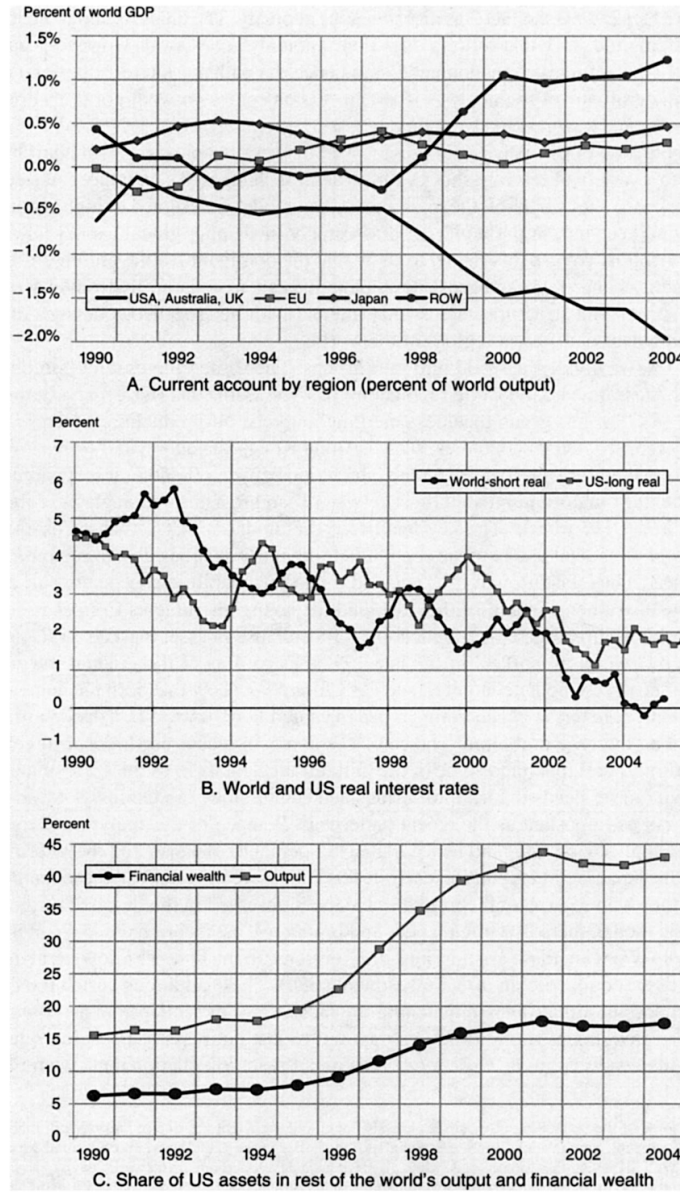


FIGURE 1. THREE STYLIZED FACTS

**Figure 1: Three Stylized Facts.** The figure plots current accounts by region, world and U.S. real interest rates, and the share of U.S. assets in the rest of the world's portfolios.

## 8.2 Figure 2: The Metzler Diagram and Figure 3: The Metzler Diagram for a Permanent Drop in $\delta^R$

**Read:** Figure 2 shows the asset supply and demand diagram. Figure 3 applies the diagram to a permanent decline in  $R$ 's asset-producing capacity.

**Economic message:** A fall in  $\delta^R$  shifts asset supply inward in  $R$ , lowers the world interest rate, and redirects savings toward  $U$  assets.



**Interpretation:** Figure 2 is the workhorse intuition: equilibrium interest rates clear asset supply and demand. Figure 3 shows how an asset-supply collapse in one region can generate a current account deficit in the asset-producing region by pushing global savers into its assets.

FROM MODEL OF GLOBAL IMBALANCES

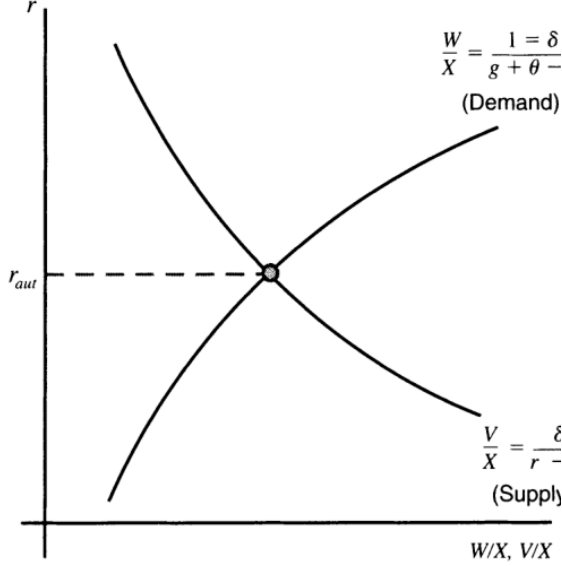


FIGURE 2. THE METZLER DIAGRAM

(a) Figure 2: The Metzler Diagram.

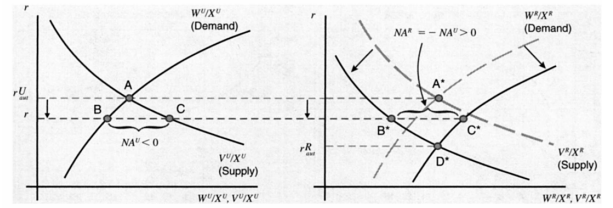


FIGURE 3. THE METZLER DIAGRAM FOR A PERMANENT DROP IN  $\delta^R$

(b) Figure 3: Permanent drop in  $\delta^R$ .

**Figures 2 and 3.** The first figure explains equilibrium asset-market clearing; the second shows how a collapse in foreign asset supply shifts the world equilibrium.

### 8.3 Figure 4: Permanent Collapse in $\delta^R$ and Table 1: Main Parameters

**Read:** Figure 4 plots the dynamic response to a permanent collapse in  $R$ 's asset-producing capacity. Table 1 reports the calibration parameters.

**Economic message:** The simulation produces a persistent current account deficit, a lower real interest rate, a worsening net foreign asset position, and a rising global portfolio share of  $U$  assets.

**Interpretation:** Table 1 shows that the mechanism is generated with a small parameter set. Figure 4 shows the dynamic mechanism in action: the asset-producing region borrows persistently because the rest of the world demands its assets.

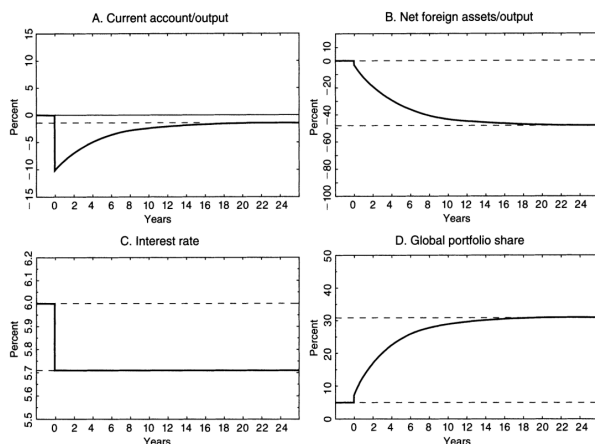
FIGURE 4. PERMANENT COLLAPSE IN  $\delta^R$ (a) Figure 4: Permanent collapse in  $\delta^R$ .

TABLE 1—MAIN PARAMETERS

| Parameter | $\theta$ | $g$  | $\delta$ | $\lambda_0^R$ | $\mu_0^{RU}$ | $NA_0^U$ | $\delta^R$ | tio |
|-----------|----------|------|----------|---------------|--------------|----------|------------|-----|
| Value     | 0.25     | 0.03 | 0.12     | 0.30          | 0.05         | 0        | 0.08       | pa  |

(b) Table 1: Main parameters.

**Figure 4 and Table 1.** The table summarizes the calibration; the figure shows the model's dynamic response to an asset-supply collapse.

#### 8.4 Table 2: Calibrated Exercise 1: Asian Shocks and Table 3: Calibrated Exercise 2: Emerging Trends

**Read:** Table 2 reports average values from the Asian shock experiment. Table 3 reports the emerging trends experiment.

**Economic message:** Both exercises push in the same direction: lower real rates, higher demand for U.S.-type assets, and persistent U.S.-centered current account deficits.

**Interpretation:** The Asian shock experiment emphasizes a collapse in foreign asset supply; the emerging trends experiment emphasizes increasing foreign asset demand from fast-growing emerging markets. The two exercises are complements rather than substitutes.

TABLE 2—CALIBRATED EXERCISE 1: ASIAN SHOCKS

| Interval (years)     | Model     |       |           |       |           |       |           |       |          |       | Data                |       |
|----------------------|-----------|-------|-----------|-------|-----------|-------|-----------|-------|----------|-------|---------------------|-------|
|                      | 1990–1997 |       | 1997–2006 |       | 2006–2020 |       | 2020–2050 |       | $\infty$ |       | 1990–1997 1997–2006 |       |
| Shock:               | P         | T     | P         | T     | P         | T     | P         | T     | P        | T     |                     |       |
| $CA^U/X^U$           | -6.2      | -6.2  | -7.0      | -6.6  | -3.3      | -2.1  | -2.9      | -1.3  | -2.9     | -1.8  | -1.1                | -4.3  |
| $NFA^U/X^U$          | -22.3     | -22.3 | -73.4     | -72.5 | -93.7     | -83.9 | -96.2     | -64.5 | -96.3    | -59.1 | -3.9                | -17.7 |
| $r - r_{\text{int}}$ | 0.10      | 0.10  | -0.48     | -0.64 | -0.52     | -0.67 | -0.53     | -0.47 | -0.53    | -0.34 | -0.78               | -1.63 |
| $\mu^U$              | 9.7       | 9.7   | 20.0      | 19.7  | 23.3      | 21.2  | 23.7      | 17.4  | 23.7     | 16.4  | 7.3                 | 16.0  |

Notes: All variables in percent. Columns labelled "P" ("T") for permanent (transitory) shocks.

(a) Table 2: Calibrated Exercise 1: Asian Shocks.

TABLE 3—CALIBRATED EXERCISE 2: EMERGING TRENDS

| Interval (years)     | Model     |       |           |       |           |       |           |       |          |       | Data                |       |
|----------------------|-----------|-------|-----------|-------|-----------|-------|-----------|-------|----------|-------|---------------------|-------|
|                      | 1990–1997 |       | 1997–2006 |       | 2006–2020 |       | 2020–2050 |       | $\infty$ |       | 1990–1997 1997–2006 |       |
| Shock:               | P         | T     | P         | T     | P         | T     | P         | T     | P        | T     |                     |       |
| $CA^U/X^U$           | -4.2      | -4.1  | -2.1      | -2.1  | -1.6      | -1.6  | -1.7      | -1.7  | -2.7     | -1.5  | -1.1                | -4.3  |
| $NFA^U/X^U$          | -15.5     | -15.3 | -32.1     | -31.7 | -39.5     | -38.9 | -47.2     | -45.9 | -88.9    | -51.3 | -3.9                | -17.7 |
| $r - r_{\text{int}}$ | -0.01     | -0.01 | -0.13     | -0.13 | -0.18     | -0.18 | -0.23     | -0.23 | -0.50    | -0.31 | -0.78               | -1.63 |
| $\mu^U$              | 5.9       | 5.9   | 8.9       | 8.8   | 9.3       | 9.2   | 8.6       | 8.4   | 0.0      | 7.8   | 7.3                 | 16.0  |

Notes: All variables in percent. "P" ("T") for permanent (transitory) shocks.

(b) Table 3: Calibrated Exercise 2: Emerging Trends.

**Tables 2 and 3.** The calibration exercises compare model-implied averages with observed data and show the direction and approximate magnitude of global imbalances.

#### 8.5 Figure 5: Collapse in $\delta^R$ with and without FDI and Figure 6: Collapse in $\delta^R$ in the Two-Good Model

**Read:** Figure 5 shows how FDI affects the adjustment path. Figure 6 adds the real exchange rate and two-good structure.

**Economic message:** FDI and exchange-rate adjustment modify the path of the imbalance, but the core asset-scarcity mechanism remains. Financial asset shortages continue to generate low rates and demand for  $U$  assets.

**Interpretation:** FDI can reduce the long-run trade imbalance by enabling the asset-producing region to generate assets from foreign investment opportunities. The two-good model shows that exchange-rate rebalancing is incomplete because the fundamental driver is an asset-market imbalance, not just a goods-market imbalance.

380

THE AMERICAN ECONOMIC REVIEW

MARCH 2008

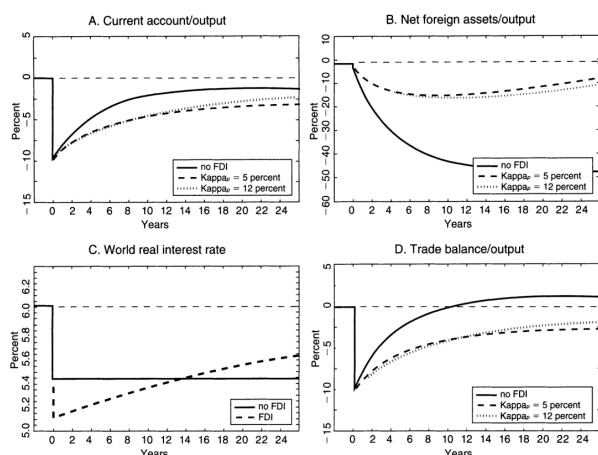


FIGURE 5. COLLAPSE IN  $\delta^R$  WITH AND WITHOUT FDI

(a) Figure 5: Collapse in  $\delta^R$  with and without FDI.

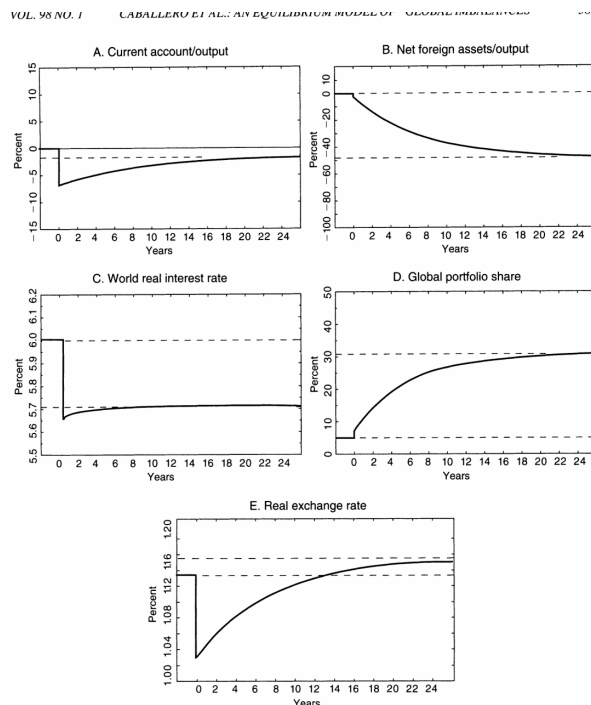


FIGURE 6. COLLAPSE IN  $\delta^R$  IN THE TWO-GOOD MODEL

(b) Figure 6: Collapse in  $\delta^R$  in the Two-Good Model.

**Figures 5 and 6.** These extensions show that FDI and real exchange-rate adjustment change the quantitative path but do not remove the central asset-scarcity force.

## 9 Short summary

- The paper explains global imbalances through cross-country differences in asset-creation capacity.
- Low-asset-supply countries demand assets from high-asset-supply countries.
- This demand lowers world real interest rates and increases the global portfolio share of U.S.-type assets.
- The U.S. current account deficit is the goods-market counterpart of foreign demand for U.S. financial assets.
- Asian crisis shocks reduce foreign asset supply; emerging-market growth raises foreign asset demand.
- FDI and exchange-rate adjustment dampen but do not eliminate the mechanism.
- The model is a precursor to later safe-asset-shortage interpretations of global macro-finance.

## **10 Conclusion**

### **10.1 Core conclusion**

Caballero, Farhi, and Gourinchas show that global imbalances can be an equilibrium response to a shortage of reliable financial assets outside the United States. The U.S. deficit is not simply an excess-consumption phenomenon; it is the counterpart of the rest of the world's demand for U.S. assets.

### **10.2 Economic intuition**

Countries with weak financial development, low pledgeability, or limited asset creation may grow and save, but cannot store wealth domestically in reliable claims. They therefore seek foreign assets. If the United States can produce these assets, capital flows into the United States and pushes down world interest rates.

### **10.3 Incremental contribution revisited**

The paper's incremental contribution is to connect global imbalances, low real rates, and U.S. asset dominance through one asset-scarcity mechanism. This changes the interpretation of current account deficits: they can reflect foreign demand for assets as much as domestic demand for goods.

### **10.4 Implications**

The model implies that global imbalances may persist as long as financial development remains uneven. Policies aimed only at exchange rates or goods-market adjustment may have limited effects if the deeper problem is a shortage of reliable financial assets.

### **10.5 Limitations**

The model is stylized and abstracts from many short-run frictions, institutional details, banking crises, and endogenous financial development. It is best read as a mechanism and calibration paper rather than a detailed forecasting model.

### **10.6 Takeaway**

The key takeaway is that global macro-finance must account for the supply of financial assets. The world does not simply trade goods and capital; it trades stores of value. When some regions cannot produce enough stores of value, global imbalances emerge.